PROBLEM 44

# The Little Prince Throwing Seeds

Original statement · Problems.pdf, p. 31

Angular Momentum + Vis-Viva

The little Prince is on his spherical asteroid, which has radius $R$ and mass $M$, and does not rotate. While he sits at the equator, he throws a baobab seed with an initial velocity of magnitude $v_0$ and angle $\beta = 60^\circ$ with respect to the local horizontal. Knowing that it reaches an apocenter of $r_a = 3R$ before falling back, determine $v_0$ as a function of the parameters of the problem and the universal gravitational constant $G$.

### Worked Solution (4-Phase Methodology)

PHASE 1

#### Physical Statement, Hypotheses & Parameters

- **System:** a seed of mass $m \ll M$ launched from the surface ($r_0 = R$) with speed $v_0$ at $60^\circ$ above the local horizontal; it reaches the apocenter $r_a = 3R$.
- **Hypotheses:** spherically symmetric, non-rotating asteroid, so the gravitational force $\mathbf{F} = -\dfrac{GMm}{r^2}\mathbf{e}_r$ is central and conservative with $\mu = GM$; vacuum (no drag); inertial frame $S_0$ centred on the asteroid.
- **Degrees of freedom:** free particle, planar motion by central-force conservation of $\mathbf{H}_O$; coordinates $(r, \theta)$.
- **Initial state:** $r = R$, radial component $v_{r0} = v_0 \sin 60^\circ = \frac{\sqrt{3}}{2} v_0$ and transverse component $v_{\theta 0} = v_0 \cos 60^\circ = \frac{1}{2} v_0$.

| Symbol | Meaning | SI unit |
|---|---|---|
| $M$, $R$ | mass and radius of the asteroid | kg, m |
| $G$ | universal gravitational constant, $6.6743 \times 10^{-11}$ | $\mathrm{m^3/(kg\ s^2)}$ |
| $\mu = GM$ | gravitational parameter | $\mathrm{m^3/s^2}$ |

| Symbol | Meaning | SI unit |
|---|---|---|
| $\beta$ | launch angle above the horizontal | $60^\circ$ |
| $r_a$ | apocenter radius | $3R$ |

PHASE 2

## Reference Frames, Vector Bases & Coordinate Geometry

- **Change of basis (polar basis in the orbital plane).** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis rotated by the true anomaly $\theta$ about $\mathbf{k}$ (along $\mathbf{H}_O$):

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

- **Velocity in $\mathcal{B}_1$:** $\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta$; at launch $\dot r = v_0\sin 60^\circ$ and $r\dot\theta = v_0\cos 60^\circ$. At the apocenter $\dot r = 0$, so the velocity is purely transverse.

PHASE 3

## Step-by-Step Mathematical Deduction

### 1. Conservation of specific angular momentum

**Pedagogical Justification:** The gravitational force passes through $O$, so $\mathbf{M}_O = \mathbf{0}$ and $h = \|\mathbf{r}\times\mathbf{v}\| = r^2\dot\theta$ is constant. Evaluate it at launch and at the apocenter, where $v_a = r_a\dot\theta_a$.

$$h = R\,(v_0\cos 60^\circ) = \frac{1}{2}Rv_0, \qquad h = r_a v_a = 3Rv_a \implies v_a = \frac{v_0}{6}$$

**2. Conservation of specific energy**

> **Pedagogical Justification:** Gravity is conservative with $V_0 = -\mu/r$ (derived with Barrow's rule, reference at infinity), so $\xi = \frac{1}{2}v^2 - GM/r$ takes the same value at launch and at the apocenter.

$$\frac{1}{2}v_0^2 - \frac{GM}{R} = \frac{1}{2}\left(\frac{v_0}{6}\right)^2 - \frac{GM}{3R} = \frac{v_0^2}{72} - \frac{GM}{3R}$$

**3. Solving for $v_0$**

$$\left(\frac{1}{2} - \frac{1}{72}\right)v_0^2 = \left(1 - \frac{1}{3}\right)\frac{GM}{R} \implies \frac{35}{72}v_0^2 = \frac{2}{3}\frac{GM}{R} \implies v_0^2 = \frac{2}{3}\cdot\frac{72}{35}\frac{GM}{R} = \frac{48}{35}\frac{GM}{R}$$

$$\boxed{v_0 = \sqrt{\frac{48}{35}\frac{GM}{R}}}$$

PHASE 4

## Physical Interpretation & Dimensional Verification

- **Bound orbit:** the escape speed at the surface is $v_e = \sqrt{2GM/R}$, i.e. $v_e^2 = \frac{70}{35}GM/R$. Since $\frac{48}{35} < 2$, $v_0 < v_e$ and the orbit is an ellipse: the seed falls back.
- **Orbital parameters:** $h^2 = \frac{1}{4}R^2v_0^2 = \frac{12}{35}GMR$, $p = h^2/\mu = \frac{12}{35}R$. From $r_a = p/(1-e) = 3R$: $1 - e = \frac{4}{35}$ and $e = \frac{31}{35} \approx 0.8857$. The periapsis of the full conic is $r_p = p/(1+e) = \frac{2}{11}R < R$, inside the asteroid; only the arc from the launch point to the apocenter is physical, and the seed lands before reaching the periapsis.
- **Independent check with the Binet solution.** With $u = \frac{\mu}{h^2}(1 + e\cos\theta)$ and the launch values $u(\theta_0) = 1/R$, $u'(\theta_0) = -\dot{r}/h$ (here $\dot{r} = \frac{\sqrt{3}}{2}v_0$), the eccentricity formula of the theory page gives

$$e = \frac{h^2}{\mu}\sqrt{\left(\frac{1}{R} - \frac{\mu}{h^2}\right)^2 + \left(\frac{\dot r}{h}\right)^2} = \frac{12R}{35}\sqrt{\frac{529}{144R^2} + \frac{3}{R^2}} = \frac{12R}{35}\cdot\frac{31}{12R} = \frac{31}{35}$$

and the launch angle follows from $\tan\beta = \dot r/(r\dot\theta) = \dfrac{R}{p}\,e\sin\theta_0$ with $\cos\theta_0 = \dfrac{p/R - 1}{e} = -\dfrac{23}{31}$, $\sin\theta_0 = \dfrac{\sqrt{432}}{31}$:

$$\tan\beta = \frac{35}{12}\cdot\frac{31}{35}\cdot\frac{\sqrt{432}}{31} = \frac{\sqrt{432}}{12} = \sqrt{3} \implies \beta = 60^\circ$$

- **Numerical check** (Python, velocity-Verlet integration of $\ddot{\mathbf r} = -\mu\mathbf r/r^3$ with $\mu = R = 1$ and $v_0 = \sqrt{48/35}$): the maximum radius reached is $2.99999999981R$, in agreement with $r_a = 3R$.
- **Dimensions:** $[v_0] = \sqrt{[G][M]/[R]} = \sqrt{\dfrac{\mathrm{m}^3}{\mathrm{kg\,s}^2}\cdot\dfrac{\mathrm{kg}}{\mathrm{m}}} = \mathrm{m/s}$.
- **Key:** the official key (Problems.pdf) has no entry for Problem 44.

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*Source: Problems.pdf (PDF page 31); no key available*